

Awards and Media Features

This document provides an overview of my awards, recognitions, and media features, as listed below.

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7NEWS Young Achiever Awards South Australia 2025 (STEM Nomination)

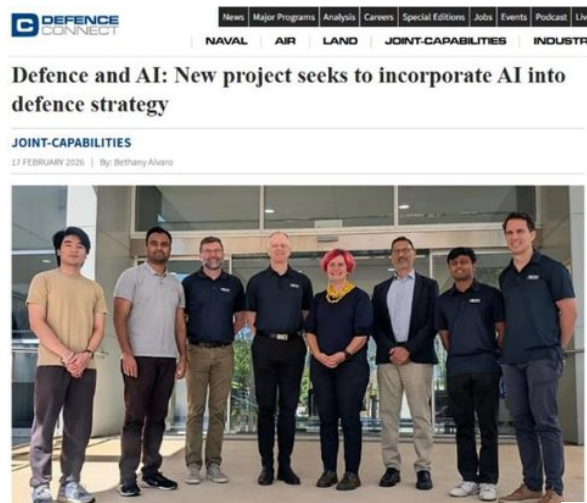
Details: <https://lnkd.in/p/gmdBgXNs>



Media Feature: AI-driven Decision-Support Research for Australia's Defence Sector



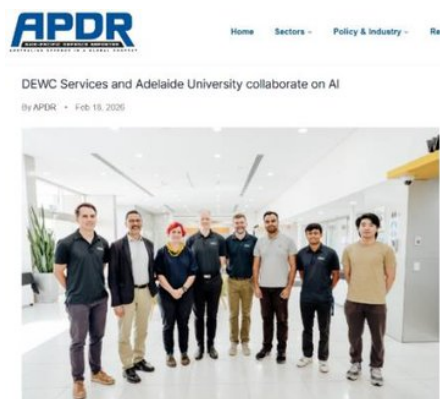
Featured by Defence Trailblazer [\[Link\]](#)



Featured by Defence Connect [\[Link\]](#)



Featured by Australian Defence Magazine [\[Link\]](#)



Featured by Asia Pacific Defence Reporter [\[Link\]](#)



Featured by Austrian Industry and Defence Network [\[Link\]](#)

Media Feature: Celebrating Our Colleagues: Dr Hussain Ahmad

Featured by: University of Adelaide (2025)

Details: <https://lnkd.in/p/gxTNA4fK>



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For all information related to the new Adelaide University – including study applications for 2026, details for commencing and current students, and Graduate Research – please visit adelaide.edu.au



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How Dr Ahmad went from bachelor's student to PhD to lecturer

Posted on Dec 2 2025 by Jo

After years of perseverance, late nights, and countless breakthroughs, Dr Hussain Ahmad recently celebrated his PhD graduation at the September 2025 ceremony. His journey from curious learner to University lecturer demonstrates that with passion and persistence, anything is possible.

What makes his story inspiring is how it began. Ahmad, who has an Electronics Engineering background, had no prior research experience. He did, however, have a keen interest in computer science and Artificial Intelligence, interests which were extended during his first role at the University in 2020 as a Cyber Security Engineer.

Soon, with support from colleagues in the Faculty of SET, Hussain's passion set the foundation for interdisciplinary research that now drives his work at the intersection of Cyber Security, Software Engineering, and Artificial Intelligence. "I was nervous at first," he says, "but my curiosity ended up shaping my entire research path."

His PhD thesis, Towards AI-Driven, Self-Adaptive, and Cyber-Resilient Software Engineering, was accepted without a single correction - the best possible outcome for any PhD candidate, and a testament to years of hard work and creativity.

Throughout his PhD, Dr Hussain's outputs were extraordinary. He:

- Led over 15 R&D projects in collaboration with industry and academic partners.
- Produced 12 research papers, including publications featured in CORE A* and A-ranked conferences and journals.
- Secured competitive research funding from both industry and academia.
- Delivered a thesis recognised as best-in-class.

Contributed to the academic community through student supervision, peer representation as a Postgraduate Researcher Representative, and advocacy for equity and inclusion as a GEDI (Gender, Equity, Diversity & Inclusion) Representative.

In recognition of his outstanding contributions, Hussain received the University's Outstanding International Student Award for research excellence, leadership, and community engagement. His creative communication also shone through his Visualise Your Thesis competition submission, where he won both the People's Choice Award and Second Prize, capturing complex research through clarity and creativity.



I try to make every class a bridge between the classroom and the real world, students shouldn't just learn - they should see where their learning takes them.

- Dr Ahmad

From a student driven by curiosity to a scholar shaping the future of cybersecurity and AI, Dr Hussain Ahmad's story is a powerful reminder that success begins not with knowing everything, but with the courage to begin, learn, and keep going.

This article was originally published in Staff News, [Celebrating our colleagues, Hussain Ahmad](#)

Tagged in [student news](#)

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Division of Academic and Student Engagement

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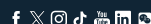
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We acknowledge the Kaurna people, the original custodians of the Adelaide Plains and the land on which the University of Adelaide's campuses at North Terrace, Waite and Roseworthy are built.

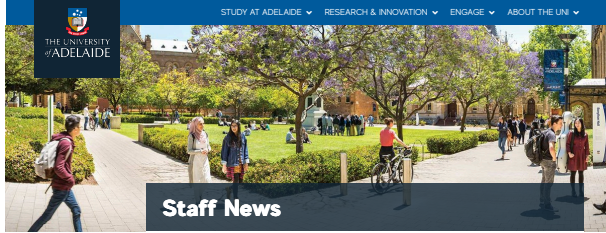


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Last updated: 2 Dec 2025

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Celebrating our colleagues: Hussain Ahmad



After years of perseverance, late nights, and countless breakthroughs, Dr Hussain Ahmad recently celebrated his PhD graduation at the September 2025 ceremony. His journey from curious learner to University lecturer demonstrates that with passion and persistence, anything is possible.

What makes his story inspiring is how it began. Ahmad, who has an Electronics Engineering background, had no prior research experience. He did, however, have a keen interest in computer science and Artificial Intelligence, interests which were extended during his first role at the University in 2020 as a Cyber Security Engineer.

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His PhD thesis, Towards AI-Driven, Self-Adaptive, and Cyber-Resilient Software Engineering, was accepted without a single correction – the best possible outcome for any PhD candidate, and a testament to years of hard work and creativity.

Throughout his PhD, Dr Hussain's outputs were extraordinary. He:

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In recognition of his outstanding contributions, Hussain received the University's [Outstanding International Student Award](#) for research excellence, leadership, and community engagement. His creative communication also shone through his [Jubilee Day Thesis competition submission](#) ², where he won both the People's Choice Award and Second Prize, capturing complex research through clarity and creativity.

"The PhD wasn't just about earning a title," he says. "It was about growing – as a person, as a thinker, and as someone who never stopped being curious."



Even before submitting his thesis, Hussain had already fulfilled another ambition, and is now a Lecturer in the School of Computer and Mathematical Sciences. "My time as a PhD candidate taught me what it feels like to be on the other side of the classroom," he says. "That perspective shaped how I teach and engage with students."

During his first semester of teaching and course coordination, Hussain quickly found his rhythm in teaching Cyber Security, Big Data, and Industry Projects, achieving an exceptional 99% overall score in the Student Experience of Learning and Teaching (SELT). Students praised his clear structure, practical examples, and dynamic teaching style, qualities rooted in his belief that clarity, simplicity, and positive energy create the most effective learning experiences.

"I try to make every class a bridge between the classroom and the real world. Students shouldn't just learn – they should see where their learning takes them."

Dr Hussain describes his teaching philosophy as rooted in five core principles:

1. Clear structure: Students value well-organised lectures and workshops that provide a logical flow, making it easier to follow concepts and retain knowledge.
2. Simplicity with depth: Breaking down complex topics into simple, digestible ideas makes learning accessible without losing substance.
3. Real-world relevance: Students like learning that connects theory to industry practices, showing how classroom concepts translate into real-world challenges and career opportunities.
4. Constructive feedback: Continuous, timely feedback helps students track progress, deepen understanding, and build confidence in their learning.
5. Positive energy: A motivating and dynamic environment keeps students engaged, inspired, and eager to participate.



This combination of innovation, teaching excellence, and community impact led to his nomination for the STEM Award at the INEWS Young Achiever Awards South Australia 2025. At the Gala Dinner Presentation at the Hilton Adelaide, where he was congratulated by Hon. Susan Close MP, Deputy Premier of South Australia, Dr Hussain shared a message that resonated with the audience:

"For me, STEM isn't just about science or technology – it's a mindset. It's about curiosity that asks, creativity that builds, and courage that transforms ideas into impact."

From a student driven by curiosity to a scholar shaping the future of cybersecurity and AI, Dr Hussain Ahmad's story is a powerful reminder that success begins not with knowing everything, but with the courage to begin, learn, and keep going.

Our graduates are the beneficiaries of a wide range of talented staff, including our educators and every year, the University showcases and celebrates our teaching excellence at the [Celebration of Learning and Teaching](#) ².

Recipients are awarded the Commendations for the Enhancement of and Innovation in Student Learning, and the Stephen Cole the Elder Awards for Excellence.

Thursday 13 November 3:00–6:00pm in The Briggs Lecture Theatre, North Terrace campus. [Register](#) ².

f x in e p

Tagged in [celebrating our colleagues](#), [feature](#), [staff news](#), [Hussain Ahmad](#)

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We acknowledge the Kaurna people, the original custodians of the Adelaide Plains and the land on which the University of Adelaide's campuses at North Terrace, Wake and Roseworthy are built.



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Outstanding International Student Award

Details: <https://lnkd.in/p/gC3hTiUm>



Media Feature: Outstanding International Student Award Recognition

Publication: Minute Mirror Newspaper (2025)

Details: [https://epaper.minutemirror.com.pk/story/title/2025/01/Page-6--\(04-01-25\)_05?new=0](https://epaper.minutemirror.com.pk/story/title/2025/01/Page-6--(04-01-25)_05?new=0)

GIKIEST graduate Hussain Ahmad makes Pakistan proud at global stage

Hussain Ahmad, a distinguished graduate of the prestigious Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, has made Pakistan proud on the global stage. Hussain has been honored with the Outstanding International Student Award at the University of Adelaide, Australia, a recognition for his exceptional achievements in research excellence, industry collaborations, and community support initiatives.

Hussain is pursuing PhD at the intersection of Cyber Security and Software Engineering, applying Machine Learning to develop innovative frameworks that enhance the security and scalability of software systems. His remarkable PhD journey includes over 10 research papers, with multiple publications in CORE A* and A-ranked venues, showcasing the global significance of his work. Hussain has collaborated with some of the world's most renowned organizations such as Schlumberger USA, CSIRO's Data61 Australia, Defence Science and Technology Group Australia, Cyber Security Cooperative Research Centre Australia, Department of State Development South Australia, Cisco Australia, Honeywell UK, Surf Life Saving South Australia, and Stone & Chalk Australia. In academia, his partnerships extend to elite institutions including Imperial College London, the University of Melbourne, Monash University, Singapore Management University, and Zayed University UAE. These collaborations have not only advanced cutting-edge research but also addressed real-world challenges, earning him a reputation as a visionary researcher and leader.

Beyond his groundbreaking research, Hussain actively contributes as a mentor, advocate, and leader in research community at the University of Adelaide. He supervises research students in cybersecurity and software engineering, guiding them to excel in their academic and professional pursuits. As the Postgraduate Researchers' Representative, Hussain advocates for the interests of researchers, fostering a positive, supportive, and constructive research environment. In his role as the Gender, Equity, Diversity, and Inclusion (GEDI) Representative, he champions initiatives that promote diverse, equitable, and inclusive environment.

His achievements serve as a powerful reminder to the youth of Pakistan that with unwavering dedication, hard work, and perseverance, no goal is too ambitious and no dream is beyond reach. They exemplify that success is not determined by one's starting point but by the determination to rise above challenges, embrace opportunities, and strive for excellence. Hussain's journey encourages young minds to believe in their potential, aim high, and work relentlessly to turn their aspirations into reality. **AGENCIES**



Media Feature: Outstanding International Student Award Recognition

Featured By: The News International (2025)

Details: <https://www.thenews.com.pk/print/1267713-distinction>

INTERNATIONAL
THE NEWS

Distinction

By PR | January 02, 2025



Ghulam Ishaq Khan Institute of Engineering Sciences and Technology building. — GIKIET website/File

Hussain Ahmad, a graduate of Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, has been honoured with the Outstanding International Student Award at the University of Adelaide, Australia, a recognition for his exceptional achievements in research excellence, industry collaborations, and community support initiatives. Hussain is pursuing PhD at the intersection of Cyber Security and Software Engineering.

99% Student Experience of Learning & Teaching Score – University of Adelaide

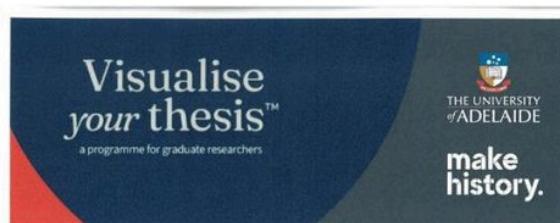
Details: <https://lnkd.in/p/gtH4Zej8>

Teacher Report for COMP SCI 7328 Concepts in Cyber Security (Hussain Ahmad)

1. Is approachable and helpful.		2. Gives clear explanations.	
Strongly Agree (52)78% 6 (14)21% 5 (1)2% Undecided (0)0% 3 (0)0% 2 (0)0% Strongly Disagree (0)0% [Total (67)] 050%100%		Strongly Agree (47)70% 6 (15)22% 5 (4)6% Undecided (1)2% 3 (0)0% 2 (0)0% Strongly Disagree (0)0% [Total (67)] 050%100%	
Statistics	Value	Statistics	Value
Mean	6.8	Mean	6.6
Median	7.0	Median	7.0
Standard Deviation	0.5	Standard Deviation	0.7
% Broad Agreement	100%	% Broad Agreement	99%
3. Makes their subject matter interesting.		4. Encourages student participation.	
Strongly Agree (43)64% 6 (18)27% 5 (5)8% Undecided (1)2% 3 (0)0% 2 (0)0% Strongly Disagree (0)0% [Total (67)] 050%100%		Strongly Agree (46)69% 6 (20)30% 5 (1)2% Undecided (0)0% 3 (0)0% 2 (0)0% Strongly Disagree (0)0% [Total (67)] 050%100%	
Statistics	Value	Statistics	Value
Mean	6.5	Mean	6.7
Median	7.0	Median	7.0
Standard Deviation	0.7	Standard Deviation	0.5
% Broad Agreement	99%	% Broad Agreement	100%
5. Provides useful and timely feedback on my work.		6. Is an effective university teacher.	
Strongly Agree (46)69% 6 (18)27% 5 (3)5% Undecided (0)0% 3 (0)0% 2 (0)0% Strongly Disagree (0)0% [Total (67)] 050%100%		Strongly Agree (47)70% 6 (18)27% 5 (1)2% Undecided (1)2% 3 (0)0% 2 (0)0% Strongly Disagree (0)0% [Total (67)] 050%100%	
Statistics	Value	Statistics	Value
Mean	6.6	Mean	6.7
Median	7.0	Median	7.0
Standard Deviation	0.6	Standard Deviation	0.6
% Broad Agreement	100%	% Broad Agreement	99%

People's Choice Award & Second Place - Visualise Your Thesis Competition

Details: [https://www.instagram.com/p/DAZkJpmy_I-/](https://www.instagram.com/p/DAZkJpmy_I/)
<https://lnkd.in/g3PQ5JK3>



The University of Adelaide
**Visualise Your Thesis
Competition 2024**

People's Choice

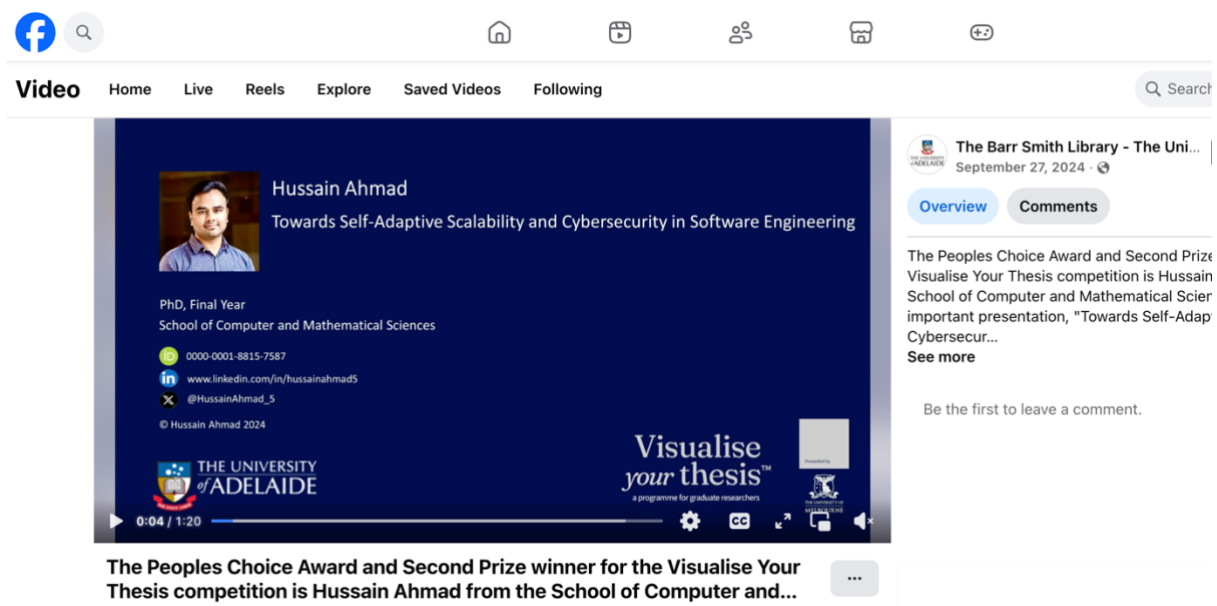
Hussain Ahmad



The University of Adelaide
**Visualise Your Thesis
Competition 2024**

Second Place

Hussain Ahmad



Media Feature: PhD Achievement in Cyber Security

Featured By: Minute Mirror Newspaper (2025)

Details: [https://epaper.minutemirror.com.pk/story/title/1759605273-3-LHR-ISBN-\(05-10-25\)_12.jpg?new=1](https://epaper.minutemirror.com.pk/story/title/1759605273-3-LHR-ISBN-(05-10-25)_12.jpg?new=1)

Dr Hussain Ahmad earns PhD in Cyber Security



Hussain Ahmad, a native of Shahdara, has completed his PhD in Cyber Security from the University of Adelaide, marking an extraordinary academic and personal milestone. His thesis was accepted without a single correction—an exceptional recognition awarded to only the most distinguished research work. During his doctorate, Dr. Ahmad led over 15 collaborative research projects with global industry and academic partners, authored 12 papers in leading international journals and conferences, and secured competitive funding to advance the field of cyber security. He also served as a mentor to students, represented his peers as a Postgraduate Researcher Representative, and promoted diversity and inclusion as a GEDI leader, earning him prestigious state and university awards. Now serving as an Assistant Professor at the University of Adelaide, Dr. Ahmad teaches Cyber Security, Big Data, and Industry Projects, while continuing to guide students in a fast-evolving digital landscape. **STAFF REPORT**

Media Feature: Research on Microservices Autoscaling and Kubernetes

Featured by: Journal of Systems and Software (JSS)

Details: <https://lnkd.in/p/gguyBBUb>



Journal of Systems and Software

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In today's fast-paced digital landscape, the cost of ineffective resource management can be staggering. 💡

[Hussain Ahmad](#), [Christoph Treude](#), [Markus Wagner](#), and [Claudia Szabo](#)

present an advance in auto-scaling for microservices with their new models, Smart HPA and ProSmart HPA. These approaches utilize reactive and proactive resource-efficient policies to optimize performance in resource-constrained environments.

Experimental results reveal that Smart HPA outperforms the Kubernetes baseline, while ProSmart HPA excels further by reducing resource misallocation and improving efficiency.

Dive into the full study here: <https://lnkd.in/eGnv65Wn>

and their replication package here: https://lnkd.in/eV2_dYwf

[#Microservices](#) [#AutoScaling](#) [#Kubernetes](#)



Journal of Systems and Software

Volume 225, July 2025, 112390



Towards resource-efficient reactive and proactive auto-scaling for microservice architectures ☆

[Hussain Ahmad](#) ^a , [Christoph Treude](#) ^b , [Markus Wagner](#) ^c , [Claudia Szabo](#) ^a

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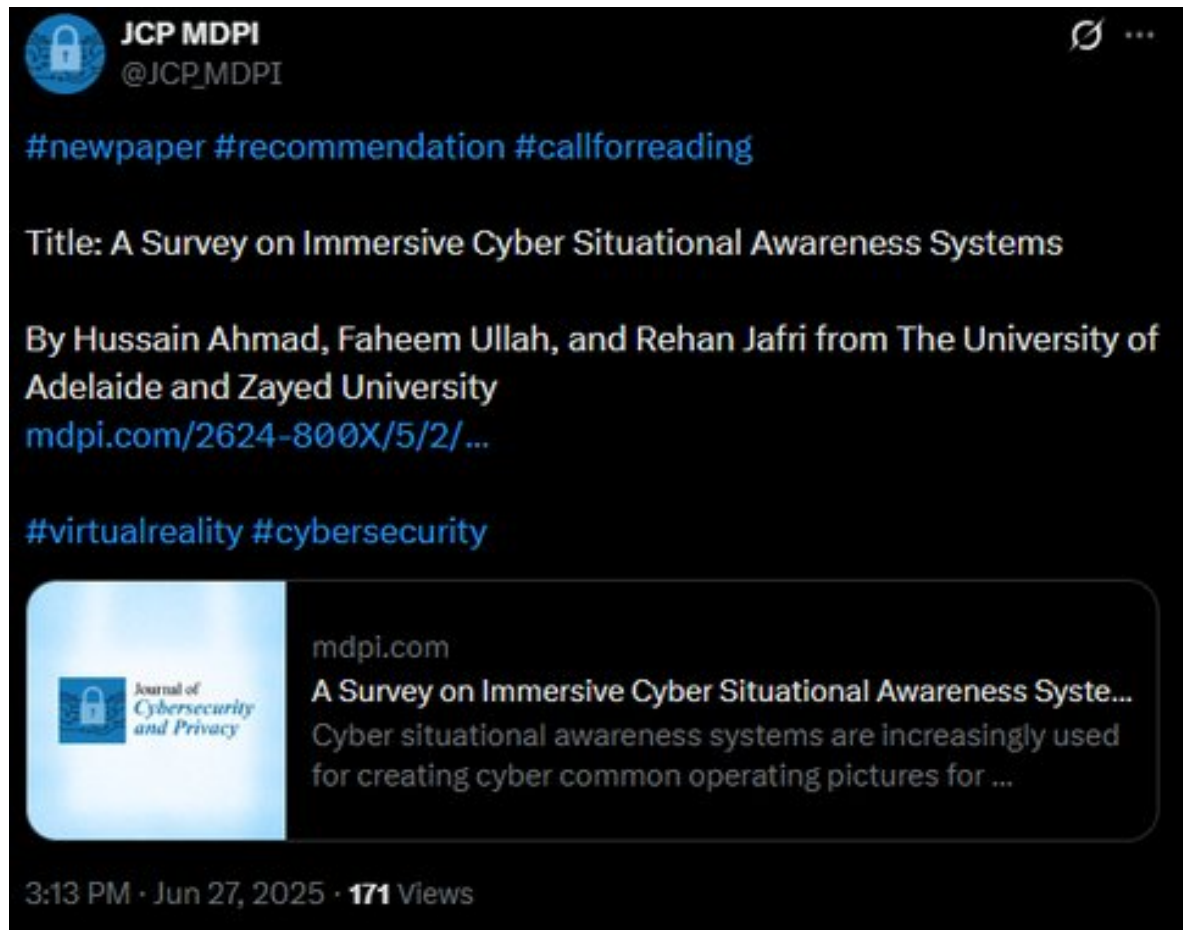
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Media Feature: Research on Cyber Situational Awareness

Featured by: Journal of Cybersecurity and Privacy, MDPI

Details: https://x.com/JCP_MDPI/status/1938473153099124885?s=20



International Conference Features



ICSA 2026 (series) / Hussain Ahmad



Registered user since Mon 4 Mar 2024

Name: Hussain Ahmad

Bio: Dr Hussain Ahmad is an Assistant Professor at the University of Adelaide, Australia. His research spans Cyber Security, Software Engineering, and Generative AI, with a strong focus on industry-driven innovation and real-world applications. He has published extensively in top-tier CORE A* and A-ranked venues and contributes to the global research community as a reviewer, program committee member, and editorial board member for leading journals and conferences. Dr Ahmad has secured multiple competitive research grants, driving breakthrough innovations that bridge academic excellence with real-world applications in critical infrastructure protection, AI-enabled systems, and software resilience. He earned his PhD at the University of Adelaide, where his interdisciplinary research at the intersection of cyber security, artificial intelligence, and software engineering laid the foundation for his ongoing contributions to Computer Science.

Country: Australia

Affiliation: University of Adelaide

Personal website: <https://www.linkedin.com/in/hussainahmad5>

X (Twitter): https://x.com/HussainAhmad_5

Research interests: Software Engineering, Cyber Security, Generative AI, Self-Adaptation

ICSA 2026 profile

Details: <https://conf.researchr.org/profile/icsa-2026/hussainahmad>



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Country: Australia

Affiliation: University of Adelaide

Personal website: <https://www.linkedin.com/in/hussainahmad5>

X (Twitter): https://x.com/HussainAhmad_5

Research interests: Software Engineering, Cyber Security, Generative AI, Self-Adaptation

APSEC 2025 profile

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ICSA 2024 (series) / Hussain Ahmad



Registered user since Mon 4 Mar 2024

Name: Hussain Ahmad

Bio: Mr Hussain Ahmad is a final-year PhD candidate at the University of Adelaide, Australia, specialising in the intersection of cyber security and scalability of software architectures. His research focuses on designing, implementing, and evaluating self-adaptive frameworks aimed at automating and optimising the security and scalability of microservices architecture. During his PhD, Hussain has collaborated on various projects with prestigious industry partners including the Cyber Security Cooperative Research Centre, CSIRO's Data61, Cisco, and Defence Science and Technology Group of Australia. Alongside his research, Hussain mentors graduate students in software engineering and cyber security, while also serving as a representative for postgraduate researchers, advocating for their interests and concerns. Hussain holds a Bachelor's degree in Electrical and Electronic Engineering.

Country: Australia

Affiliation: University of Adelaide

Personal website: <https://www.linkedin.com/in/hussainahmad5>

X (Twitter): https://x.com/HussainAhmad_5

Research interests: Software Engineering, Cyber Security, Machine Learning, Self-Adaptation

ICSA 2024 profile

Details: <https://conf.researchr.org/profile/icsa-2024/hussainahmad>

Media Article: Digital Growth vs Cyber Threats: The Critical Need for Cybersecurity

Featured By: Daily Times (2024)

Details: <https://dailytimes.com.pk/1211901/>

Digital growth vs cyber threats: The critical need for cybersecurity



By Hussain Ahmad

The rapid expansion of the internet has sparked a digital revolution in Pakistan, boosting the nation's economic and social development. However, this surge has also introduced new threats, particularly in the realm of cybersecurity. In recent years, cyber-attacks have become more frequent, sophisticated, and costly, impacting individuals, businesses, and government institutions. In 2023, a report by DAWN revealed a drastic rise in financial frauds where criminals impersonated bank employees, exploiting personal information to deceive victims. Since 40 percent of the population is uneducated, this rise in cybercrime is unsurprising yet alarming. Despite various government initiatives, limited access to primary education

remains a persistent issue in Pakistan. Many children leave school early to support their families financially, resulting in approximately 3.3 million children missing out on their formative years, basic healthcare, and education. It is unrealistic to expect these unequipped children to understand and navigate the risks associated with advanced technologies. This lack of education creates a significant barrier to comprehending and mitigating cybersecurity risks.

The pressing question now is: how can we improve cybersecurity in Pakistan? While there can be several potential solutions, one effective approach is to adopt the following three step strategy:

Understanding Cyberthreats: As the saying goes, "A problem well-defined is a problem half-solved." A critical step in mitigating cybercrime is thoroughly understanding and analysing the sources of threats. By identifying where risks originate, more effective prevention and response strategies can be developed. This involves continuous monitoring of cyber threats, analysing attack patterns, and understanding the motivations and methods of cybercriminals.

Raising Cybersecurity Awareness: Increasing public awareness about cybersecurity is essential. Education and training programs can equip individuals with knowledge about basic cybersecurity

practices and how to protect themselves against cyber threats. Cybersecurity courses covering fundamental concepts like phishing scams, identity theft, and online fraud should be made compulsory in schools. Additionally, educating businesses and individuals on best practices, such as regular software updates and data backups, is crucial.

Building Cyber Resilience: Building cyber resilience requires a multifaceted approach that includes the development and implementation of robust policies, standards, data governance, and risk management frameworks. A comprehensive cybersecurity strategy is essential for accurately identifying, assessing, and mitigating emerging threats associated with advanced technologies. Enhancing both digital and operational resilience is vital to protecting Pakistan's digital infrastructure and ensuring its economic future.

The digital transformation of Pakistan brings significant opportunities for growth and development. However, it also necessitates a concerted effort to address the rising cybersecurity threats. By understanding cyber threats, raising awareness, and building resilience, Pakistan can create a secure digital environment that supports its economic and social aspirations.

The writer is a cyber-security researcher at the University of Adelaide, Australia.

Media Article: The risks of AI dependence for developing nations

Featured By: The News International (2025)

Details: <https://e.thenews.com.pk/detail?id=382255>

The risks of AI dependence for developing nations

Hussain Ahmad

AS Artificial Intelligence (AI) reshapes technology worldwide, developing countries must carefully navigate its adoption, especially generative AI. While these AI-driven tools enhance efficiency and foster innovation, they also raise serious concerns about data sovereignty and national security.

The rise of generative AI models like ChatGPT has made advanced AI accessible to people in developing nations, unlocking new opportunities in education, business, and governance. However, this rapid and widespread adoption is like a virus - once it spreads, it becomes nearly impossible to regulate. The challenge is not just about access but control: How can a country restrict the use of AI models when their ownership, governance, and data processing are controlled by developed nations? This dependency raises serious concerns about data sovereignty, national security, and economic stability. If developing nations rely on foreign AI infrastructure, they risk

losing control over critical information flows, making them vulnerable to external influence. The ability of AI providers to dictate terms, restrict access, or shape information ecosystems could deepen inequalities, effectively placing developing nations in a position of digital dependence. Critics argue that this dependency could evolve into a new form of digital colonialism, where developing nations become increasingly reliant on AI infrastructure, they neither own nor control.

The integration of AI into daily life marks an irreversible shift in human behavior and societal function. As AI tools become embedded in everyday decision-making, from personal choices to government operations, this dependence becomes neurologically and structurally hardwired, much like how smartphones transformed from luxury to necessity. For developing nations, this behavioral lock-in poses an unprecedented sovereignty challenge. When a population's cognitive patterns and daily operations become dependent on for-

eign-controlled AI systems, national autonomy is compromised at its most fundamental level. The sovereignty threat extends beyond data and infrastructure. It reaches into the collective thinking and decision-making capabilities of entire populations, making traditional notions of national independence increasingly fragile.

To navigate these challenges, developing nations must weigh the risks against the opportunities. On one hand, restricting foreign AI usage can help protect national data and reduce reliance on external entities. However, this approach risks stifling innovation and falling behind in the global AI race. On the other hand, investing in domestic AI capabilities offers greater sovereignty and control but requires significant resources such as advanced infrastructure, skilled talent, and sustained research. Several countries have already begun initiatives to build local AI research centers and develop indigenous models, though the gap with leading AI nations remains

substantial.

The real challenge lies in striking a balance: leveraging foreign AI for immediate gains while steadily building the foundations for long-term technological independence. Building domestic AI capabilities is not just about technological independence but it is about ensuring long-term economic competitiveness. The initial investment may be significant, but the alternative is perpetual technological dependence.

In an AI-driven future, technological self-reliance is becoming as crucial to national sovereignty as traditional measures of self-determination. The cost of investing in domestic AI capabilities is high, but the cost of dependence could be even greater, potentially restricting economic and strategic autonomy for decades to come. Developing nations must act now, striking a careful balance between adoption and self-sufficiency to secure their place in the global AI landscape. (The writer is assistant professor of cybersecurity and software engineering at The University of Adelaide, Australia)

Media Article: Importance of cybersecurity education in Pakistan

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Importance of cybersecurity education in Pakistan

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In 2023, Pakistan experienced a significant 17 percent increase in cyberattacks compared to the previous year, highlighting a troubling trend in the nation's cybersecurity landscape.

With 40 percent of the population uneducated, this rise in cybercrime is both expected and alarming. As technology advances, cybersecurity becomes increasingly vital for individuals and organizations.

The ever-evolving nature of cyber threats necessitates a comprehensive understanding of online safety, making cybersecurity education a critical priority for Pakistan.

Integrating cybersecurity education into the school curriculum is essential to address growing challenges. Early education in cybersecurity helps students navigate the digital world safely and responsibly.

Learning digital ethics is crucial, teaching students about civic responsibilities and ethical behavior, such as respecting online privacy and safely using social media.

Starting this education early ensures that stu-

dents develop good digital habits and maintain a safe online presence.

Teaching cybersecurity in schools equips them to handle the risks associated with digital tools and platforms.

Protecting personal information online is paramount; students need to learn about strong passwords, recognizing phishing attempts, and understanding privacy settings to safeguard themselves from identity theft and other cybercrimes.

Cybersecurity education can also help prevent cyberbullying by promoting awareness of online behavior and its potential consequences. This knowledge contributes to creating a safer online environment.

Moreover, cybersecurity skills are increasingly valuable in the job market, preparing students for future careers in various fields.

However, implementing cybersecurity education in Pakistan presents several challenges. There is a shortage of skilled educators with contemporary digital knowledge and expertise in cybersecurity.

Training teachers and updating the curriculum requires significant investment and continuous revi-

sion to keep pace with evolving threats.

Awareness and emphasis on cybersecurity at the policy level are lacking, complicating integration into the national education framework.

Many schools, especially in rural areas, lack the necessary infrastructure and access to technology to effectively teach cybersecurity.

Socio-economic disparities mean not all students have equal access to digital devices and the internet, creating a gap in the effectiveness of cybersecurity education.

Despite these challenges, prioritizing cybersecurity education is essential.

By investing in teacher training, updating curricula, and ensuring equitable access to technology, Pakistan can equip its next generation with the skills and knowledge to navigate the digital world safely and responsibly.

This investment will not only protect individuals but also strengthen national security and promote economic growth in the increasingly digital global economy.

(The writer is a Cybersecurity Researcher at The University of Adelaide, Australia)

Award: Award of Appreciation for Serving as a Judge at the Robotics Competition
By: National Science and Engineering Fair (2017)

